HEALTHCARE INFORMATION SYSTEMS PROGRAM (HCIS)

Mission

The mission of the Healthcare Information Systems program is devoted to providing students with a state-of-the-art curriculum, maintaining persistent relationships with industry to create opportunities for students, cultivating learners with a sense of responsibility towards the profession and a passion for lifelong learning, and providing the educational experiences that will enable our students to be innovative leaders.

Objectives

The educational objectives of the Healthcare Information Systems program intend to enable its graduates, within a few years of graduation, to:

- Solve healthcare computing problems that meet local, regional, or global markets.
- Demonstrate professional skills and ethical behavior in practicing their profession
- Progress in computing and other professionally related fields at the level of professional employment or graduate studies

Program Outcomes

Upon successful completion of the BS degree in Healthcare Information Systems, graduates should demonstrate:

a. An ability to apply knowledge of health information systems and use current technologies, techniques, skills, and tools necessary for effective delivery of healthcare services appropriate to healthcare organizations;
b. An ability to apply health analytic methods to influence the planning, organizing, administering, and policy creation of healthcare organizations;
c. An ability to design, implement, and evaluate healthcare systems, processes, and procedures to meet desired needs;
d. An ability to evaluate healthcare processes, practices, management of data, technologies, and conditions to ensure compliance with established quality and professional standards;
e. An ability to function effectively on teams to accomplish a common goal;
f. An understanding of professional, ethical, legal, security, and social issues and responsibilities;
g. An ability to communicate effectively with a range of audiences.
h. Recognition of the need for and an ability to engage in continuing professional development.

Career Opportunities

With a Healthcare Information Systems degree, students will gain foundational knowledge to develop and apply information technologies and systems to meet the growing needs of healthcare organizations. Given the importance of information accuracy, privacy, and security, HCIS graduates will be prepared for involvement in regulatory compliance and quality assessment activities designed to ensure that health information systems support patient care and safety. HCIS graduates can work with nurses, physicians, other healthcare providers, managers, and technical specialists in a variety of settings such as hospitals, long-term-care facilities, insurance and managed care organizations, government agencies, and vendor firms.

Program Overview

RHU's Healthcare Information Systems (HCIS) program combines concepts from healthcare, information technology, and leadership practice to prepare graduates with the knowledge and skills needed to be effective designers and developers of health information systems. The HCIS program at RHU develops students' computational and critical thinking skills to create, not simply use, new healthcare technologies and systems. Students learn to think logically and solve problems by acquiring the fundamental principles of science and advanced techniques currently used in practical systems development. They also learn oral communication and presentational skills and how to work on individual and team projects to develop new systems that enhance time management and team coordination. RHU's HCIS program graduates are lifetime learners; they can adapt quickly to this ever-changing, challenging field.

To obtain a Bachelor of Science degree in HCIS, the student must complete 100 credit hours. These hours include University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours is shown in the following table:

BS in Healthcare Information Systems (100 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	9	–	13	9	31	31
College Requirement	–	–	–	–	–	–
Program Requirement	42	9	18	–	69	69
Credits	51	9	31	9	100	100

I. University Requirements (General Education)

The list of the University's required courses and descriptions is presented on the introductory pages of the College of Arts and Sciences section of this catalog.

II. College Requirements

The list of college-required courses and descriptions is presented in the introductory pages of the College of Arts and Sciences section of this catalog.

III. Program Requirements

A. Mandatory Requirements

Mathematics
The mathematics requirement courses are listed in the table below.

Course #	Title	Credits	Prerequisites
MATH 210	Discrete Mathematics	3	
MATH 215	Calculus III	3	
MATH 311	Linear Algebra	3	
MATH 421	Numerical Analysis	3	MATH 311

Business Requirements
The business requirement courses are listed in the table below.

Course #	Title	Credits	Prerequisites
BMGT 300	Project Management	3	Junior Standing
BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210

Major Requirements
The HCIS mandatory core courses are listed in the table below.

Course #	Title	Credits	Prerequisites
COSC 215	Advanced Programming and Data Structures	3	COSC 214
COSC 231	Database Management Systems	3	COSC 214
COSC 316	Design and Analysis of Algorithms	3	COSC 215
COSC 333	Web Programming	3	COSC 214
COSC 341	Software Engineering	3	COSC 214
COSC 360	Networking	3	COSC 214
COSC 360L	Networking Lab	1	Co-req.: COSC 360
COSC 480	Machine Learning and Data Mining	3	COSC 214; MATH 351 or BADM 250
HCIS 240	Medical Terminology	3	
HCIS 341	Health Information Systems I	3	BIOL 210
HCIS 342	Health Information Systems II	3	HCIS 341

HCIS 343	Information Security and Privacy in Healthcare	3	HCIS 341
HCIS 399	Co-op Training Experience	1	ENGL 217; Senior Standing.
HCIS 444	Healthcare Business Intelligence & Data Analysis	3	HCIS 341
HCIS 445	Research in Healthcare Information Systems	1	Senior Standing
HCIS 490	Summative Learning Experience	3	ENGL 217

B. Technical Electives

As part of the Bachelor of Science in Healthcare Information Systems (HCIS) program, students must complete 9 credit hours of technical electives. These courses allow students to focus on a specific area for in-depth knowledge and understanding. The student can also mix and match elective courses from different areas to gain a broader exposure to the various HCIS disciplines. In cooperation with the academic advisor, the student should select electives that best align with their needs and aspirations. The listed technical elective courses and other courses from engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in areas of interest.

Course #	Title	Credits	Prerequisites
BIOM 520	Health-care Facility Management	3	HCIS 341
BIOM 565	Introduction to E-Healthcare	3	HCIS 341
COSC 434	Advanced Web Programming	3	COSC 231 and COSC 333
COSC 435	Mobile Application Development	3	COSC 214; Senior Standing
COSC 436	Game Programming	3	COSC 214
COSC 438	Introduction to Software Testing	3	COSC 341
COSC 451	Operating Systems	3	COSC 214
COSC 461	Advanced Networking	3	COSC 360 or CCEE 354
COSC 463	Cybersecurity Operations	3	COSC 360
COSC 481	Artificial Intelligence	3	COSC 214
COSC 482	Data Science and Web Scraping	3	Co-req.: COSC 333 and MATH 351
COSC 498	Special Topics	3	
HCIS 498	Special Topics	3	Senior Standing
CCEE 561	Computer Vision	3	COSC 214
CCEE 566	Natural Language Processing	3	COSC 214 or COSC 214.

* The list of elective courses is subject to change.

In addition, the students can take the following three labs instead of one elective.

Course #	Title	Credits	Prerequisites
Networking			
COSC 460L	Networking Lab 2	1	Co-req.: COSC 360L
COSC 461L	Networking Lab 3	1	COSC 460L
COSC 462L	Networking Lab 4	1	Co-req.: COSC 461L or CCEE 555L.

C. Summative Learning Project

Students must complete a 3-credit-hour course in which they work, preferably in groups, on a problem of concern to industry or the community at large, or on a promising idea to innovate. The SLP could be an extension of the projects students worked on in the Engineering for the Community course, bringing them to a more useful outcome.

D. Co-op Training Experience

Each student must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member is assigned to mentor the student and follow up on their progress throughout the Co-op period by conducting field visits and ensuring that the student's performance aligns with their aspirations and the employer's needs. Students must submit a formal report and a poster and make a formal presentation about their Co-op experience. While ENGL 217 and Senior Standing are prerequisites for co-op training, the program must stipulate additional conditions.

Course Coding

Code numbers designate the courses offered in the HCIS program in the form of (HCIS abc), where:

a	Designate the year (level)
b	Designate the focus area as follows: 1. Software Engineering and Quality Assurance; 2. Network Administration and Security; 3. Healthcare
c	Designate the course sequence in a focus area.

Study Plan

The Bachelor of Science in Healthcare Information Systems encompasses 100 credit hours over six semesters and two summer sessions. The following study plan serves as a roadmap for smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (15 Credits)			
ARAB 212	Arabic Language and Communication	2	

BMKA 200	Introduction to Marketing	3	Co-req.: ENGL 210
	Community & Sustainability	1	Co-req.: ENGL 210
COSC 214	Introduction to Programming	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
Year 1, Spring Semester (16 Credits)			
COSC 215	Advanced Programming and Data Structures	3	COSC 214
COSC 231	Database Management Systems	3	COSC 214
CMNS 205	First Aid	1	
ENGL 217	Professional English Communication	3	ENGL 210
HCIS 240	Medical Terminology	3	
MATH 210	Discrete Mathematics	3	
Year 1, Summer Semester (6 Credits)			
BIOL 210	Human Anatomy and Physiology and Lab	3	
	Social Science/Culture Elective I	3	
Year 2, Fall Semester (16 Credits)			
COSC 316	Design and Analysis of Algorithms	3	COSC 215
COSC 333	Web Programming	3	COSC 214
COSC 360	Networking	3	COSC 214
COSC 360L	Networking Lab	1	Co-req.: COSC 360
HCIS 341	Health Information Systems I	3	
MATH 351	Probability and Statistics	3	MATH 215
Year 2, Spring Semester (15 Credits)			
COSC 341	Software Engineering	3	COSC 214
HCIS 342	Health Information Systems II	3	HCIS 341
HCIS 343	Information Security and Privacy in Healthcare	3	HCIS 341
MATH 311	Linear Algebra	3	
	HCIS Major Elective I	3	
Year 2, Summer Semester (1 Credit)			
HCIS 399	Co-op Training Experience	1	ENGL 217; Senior Standing
Year 3, Fall Semester (15 Credits)			
MATH 421	Numerical Analysis	3	MATH 311
GRDS 440	Design and Social Impact	3	
HCIS 444	Healthcare Business Intelligence and Data Analysis	3	HCIS 341
	Humanities/Fine Arts Elective	3	
	HCIS Major Elective II	3	
Year 3, Spring Semester (16 Credits)			
BMGT 300	Project Management	3	Junior Standing

COSC 480	Machine Learning and Data Mining	3	COSC 214; MATH 351 or BADM 250
HCIS 490	Summative Learning Experience	3	ENGL 217
HCIS 445	Research in Healthcare Information Systems	1	Senior Standing
	HCIS Major Elective III	3	
	Social Sciences/Culture Elective II	3	

Courses Description

I. Mandatory Courses

Descriptions of the major mandatory courses are given below.

COSC 214	**Introduction to Programming**	**3(2,2)**
This course presents the fundamentals of structured and modular programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to CCEE 214.		

COSC 215	**Advanced Programming and Data Structures**	**3(2,2)**
Continuing COSC 214, this course covers the basics of object-oriented programming (OOP) languages and data structures. It covers pointers, classes, encapsulation, constructors/destructors, object instantiation, and templates. The course also details data structures and applications such as stacks, queues, lists, and trees. **Prerequisite***:* COSC 214. Equivalent to CCEE 216.		

COSC 231	**Database Management Systems**	**3(3,0)**
This course exposes students to the fundamental concepts for designing, using, and implementing database systems. It covers database concepts, data modeling, relations, normalization, and data manipulation languages. **Prerequisite***:* COSC 214. Equivalent to CCEE 315.		

COSC 316	**Design and Analysis of Algorithms**	**3(3,0)**
This course consolidates algorithm design and programming techniques. It provides an extended study of object-oriented programming properties, data structures, and data abstraction, as well as an introduction to complexity considerations. **Prerequisite***:* COSC 215.		

COSC 333	**Web Programming**	**3(3,0)**
This course teaches students how to develop and implement web-based programs, with an emphasis on front-end programming. It introduces students to web development, client-side languages, and the styles needed to build functional, responsive websites. The course covers HTML5, CSS3, JavaScript/jQuery, and responsive design. **Prerequisite***:* COSC 214. Equivalent to CCEE 411.		

COSC 341	Software Engineering	3(3,0)
This course provides a deep and comprehensive understanding of the fundamentals of software engineering. The course introduces students to software process models, project management, software requirements, and engineering processes. The course covers SDLC stages, UML diagrams, and the use of CASE tools. **Prerequisite***:* COSC 214. Equivalent to CCEE 310.		

COSC 360	Networking	3(3,0)
This course enables students to gain fundamental knowledge of computer networks, appreciate various tradeoffs and choices in networking, learn to design and analyze protocols, and prepare to study advanced topics in the field of networks. **Prerequisite***:* COSC 214. Equivalent to CCEE 354.		

COSC 360L	Networking Lab	1(0,2)
This Lab helps prepare students seeking to pass advanced Cisco Certifications. The student will acquire knowledge of network equipment and protocol functionality, how to build a simple Ethernet network using routers, switches, and computers, and how to use router CLI commands to perform basic configuration and verification. The student will also learn valuable network problem-solving techniques and concepts. This course is an accompanying Lab for COSC 360. **Co-requisite**: COSC 360 or CCEE 354. Equivalent to CCEE 354L.		

COSC 480	Machine Learning and Data Mining	3(3,0)
This course introduces students to the basic knowledge representation and learning techniques. The emphasis is on understanding the data mining process, applying appropriate approaches to solve practical problems, and developing intelligent software applications. The course covers several topics about classification, prediction, and clustering. **Prerequisites**: COSC 214; MATH 351 or BADM 250. Equivalent to CCEE 564.		

HCIS 240	Medical Terminology	3(3,0)
This course introduces elements of medical terminology, including the foundations of words used to describe the human body and its conditions, terminology for medical procedures, and the names of commonly prescribed medications. Spelling, pronunciation, and meanings of terms used in a professional healthcare system are covered, as well as the recognition of common abbreviations. **Prerequisite**: None.		

HCIS 341	Health Information Systems I	3(3,0)
This course introduces healthcare, medical, and business processes from a software design perspective. Topics include the history of – and current topics related to – the healthcare delivery process, healthcare functions supported by hospital IT departments, and interactions between healthcare and business data domains, as well as between medical and allied health professionals.		